Student Success Predictor: A Comparative Study of Machine Learning Algorithms for Early Academic Risk Prediction

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Abstract

Early identification of academically at-risk students allows schools to allocate support before failure occurs, but deploying the strongest possible predictive model is not the real constraint: grades themselves become available only late in the course. This project compares three standard classifiers - Logistic Regression, Decision Tree, and Random Forest - on the UCI Student Performance dataset (395 secondary-school students of Mathematics) under an honest early-prediction protocol: no grade feature (G1, G2, or G3) is available to the models. The binary target is defined by a project-level operational rule (at risk if G3 < 10 on the 0-20 Portuguese scale). A stratified 80/20 split (seed 42) is held out, while tuning uses 5-fold cross-validation on the training split only. With the at-risk class as positive, the measured test results were: Random Forest: Accuracy 0.696, Precision 0.542, Recall 0.500, F1 0.520, ROC-AUC 0.644; Logistic Regression: Accuracy 0.582, Precision 0.387, Recall 0.462, F1 0.421, ROC-AUC 0.594; Decision Tree: Accuracy 0.582, Precision 0.387, Recall 0.462, F1 0.421, ROC-AUC 0.567. Random Forest achieved the best F1 and ROC-AUC, yet all models remain far from perfect; notably, including the intermediate grades G1/G2 in a separate, non-comparable experiment raises test accuracy from roughly 0.7 to 0.87 - quantifying how much late-course information 'leaks' into the task when it is allowed. We report confusion matrices, ROC curves, permutation importance, class imbalance effects (class_weight='balanced'), and discuss limitations: small sample size, single-context cohort, proxy-based feature availability, and the fact that findings are predictive associations, not causal effects. The trained pipeline and an interactive, disclaimer-first Streamlit application are released as an educational prototype with reproducible configuration and tests.

Keywords

Educational data mining; machine learning; classification; student performance prediction; data leakage; imbalance; scikit-learn; Streamlit

1. Introduction

Academic failure rarely starts at the final exam. Schools that can identify students at risk early enough to intervene have a practical advantage, which is why machine learning on educational records (Educational Data Mining; Romero & Ventura, 2010) has grown as a research area. The UCI Student Performance dataset (Cortez & Silva, 2008; Cortez, 2008) is a standard, openly licensed testbed for this problem and is the dataset used here (Cortez, 2008, DOI 10.24432/C5TG7T).

A recurring pitfall in this setting is data leakage: models are often trained with intermediate grades (G1, G2) that be available only near the end of the course, making them excellent predictors of the final grade (G3) but useless for an early system (Cortez, 2008, explicitly notes the strong correlation between G3 and G1/G2). This project therefore deliberately excludes all grade columns from the feature set in the primary experiment and treats grade-free prediction as the honest scenario.

This project - executed by a two-student team as part of a university data science course - contributes (i) a leakage-aware, reproducible experimental design; (ii) a controlled comparison of three classic algorithms; (iii) honest reporting of measured metrics including their uncertainty; and (iv) an interactive application that surfaces the measured results and clearly discloses their limits.

2. Problem Statement

Can a machine learning classifier, built on student attributes that are available before grades are known (family, demographic, social, and school-context features), identify students who will end the course below the pass threshold? How do three representative algorithms compare on this task under stratified, leakage-safe, reproducible conditions?

3. Research Objectives

1. 1. Define an explicit, documented target: at_risk = G3 < 10 on the dataset's 0-20 grading scale.
2. 2. Prevent data leakage by excluding G3 (defines the target) and G1/G2 (unavailable early) in the primary experiment.
3. 3. Train, tune (training data only) and evaluate Logistic Regression, Decision Tree, and Random Forest under identical conditions.
4. 4. Report Accuracy, Precision, Recall, F1, confusion matrix, ROC-AUC and training time on a held-out test split.
5. 5. Quantify the impact of including intermediate grades in a clearly separated experiment.
6. 6. Deliver an interactive application that only shows measured results.

4. Research Questions

RQ1. How do the three algorithms compare when predicting an explicitly defined academic risk category from attributes available before the prediction time?

RQ2. How large is the performance gap between the leakage-free early scenario and the intermediate-grades (midterm) scenario, and why is it not a fair comparison?

RQ3. Which evaluation metric best serves the goal of identifying students who may need support, given class imbalance?

5. Literature Review

Cortez and Silva (2008) introduced the dataset used here and modeled both Mathematics and Portuguese course performance under binary/five-level and regression tasks; they predict G3 (or binarized G3) with techniques including decision trees, random forests, neural networks and SVMs, and explicitly observe that excluding G1/G2 makes the task harder but more useful for early prediction. In the broader educational data mining literature, Romero and Ventura (2010) survey methods, tools and evaluation practice for predicting academic outcomes.

On algorithms: regularized logistic regression is a standard linear baseline for binary classification (Cox, 1958; Hastie, Tibshirani, & Friedman, 2009). Decision trees partition the feature space with axis-aligned rules (Breiman et al., 1984), are interpretable but prone to overfitting without pruning. Random forests (Breiman, 2001) average many decorrelated trees to reduce variance and typically improve generalization on small tabular datasets such as this one.

On evaluation: overall accuracy is known to be a weak indicator under class imbalance (He & Garcia, 2009); recall of the minority class determines how many at-risk students are actually found, precision controls false alarms, and F1 balances the two. Precision-recall style evaluation is being noted as more informative than ROC on imbalanced data in some settings (Saito & Rehmsmeier, 2015), which is why this project reports the full confusion matrix alongside scalar metrics.

6. Dataset Description

Source: UCI Machine Learning Repository - Student Performance (Cortez, 2008), DOI 10.24432/C5TG7T, license CC BY 4.0. The primary file is student-mat.csv (Mathematics): 395 students, 33 columns, 0 missing values. A Portuguese-language file (student-por.csv, 649 students) is kept for extensions but not used in the reported experiment.

The 33 columns cover school, demographic (e.g. sex, age, address, family size/education/jobs), behavioural (study time, failures, absences), social (free time, going out, alcohol use), and grade

variables G1-G3. Full definitions are stored in src/config.py and surfaced in the application's Methodology page.

The target G3 distribution is imbalanced: not_at_risk 265 (67%), at_risk 130 (33%) - 'at risk' is the minority class, motivating class_weight='balanced' in all models.

7. Methodology

The experiment follows a fixed, scriptable pipeline (src/train.py): load and validate the raw file; build the target; audit exclusions; stratified 80/20 split with seed 42; grid-search tuning with 5-fold stratified CV on the training split only; final evaluation on the held-out split. The test split is used once, for the reported comparison, and never for tuning or threshold selection.

8. Data Preprocessing

Numeric features (13 of 30) receive median imputation followed by standardization; categorical features (17 of 30) receive most-frequent imputation and one-hot encoding with handle_unknown='ignore' (so unknown category values at prediction time do not crash the saved pipeline). All preprocessing is wrapped in a sklearn ColumnTransformer inside a Pipeline and is fitted on the training split only. The ten excluded/segregated grade columns are audited in code: G3 is excluded in every scenario - it defines the target - and G1/G2 are excluded in the primary early scenario; the audit (src/data_loader.get_feature_metadata) reports exactly which columns were excluded and why, and a test asserts the audit.

9. Algorithm Descriptions

Logistic Regression predicts the log-odds of the positive class as a linear combination of features; we use default L2 regularization with C selected from {0.1, 1, 10}. (Cox, 1958; Pedregosa et al., 2011).

Decision Tree greedily selects axis-aligned splits that maximize impurity reduction; we tune max_depth and min_samples_leaf, a light pruning for this small dataset (Breiman et al., 1984).

Random Forest bagges hundreds of decision trees with random feature subsampling (Breiman, 2001); we tune n_estimators, max_depth and min_samples_leaf. Because the at-risk class is the minority in all splits, all three models use class_weight='balanced' so that the minority class contributes proportionally to the loss.

10. Experimental Setup

Reproducible configuration (stored verbatim in results/metrics/model_metadata.json): random seed 42; stratified split 316 train / 79 test; class balance (train) {'not_at_risk': 212, 'at_risk': 104}

and (test) {'not_at_risk': 53, 'at_risk': 26}; 5-fold stratified CV on training data scored by F1 of the positive ('at_risk') class; class_weight='balanced'; Python 3.13.5, scikit-learn 1.9.1.

11. Evaluation Metrics

Accuracy = overall correct fraction - but with imbalanced classes it can hide poor minority performance (He & Garcia, 2009). Precision(at_risk) = how many flagged students are truly at risk (false alarms cost credibility and stigma). Recall(at_risk) = how many true at-risk students the model catches - the primary quantity of interest for early support. F1 harmonizes precision and recall. ROC-AUC measures ranking quality across thresholds. Confusion matrices expose TP/FP/FN/TN explicitly.

12. Results

Measured on the held-out test split (positive class = at_risk):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.696	0.542	0.500	0.520	0.644
Logistic Regression	0.582	0.387	0.462	0.421	0.594
Decision Tree	0.582	0.387	0.462	0.421	0.567

Per-model confusion matrices and per-feature permutation importance (on the test split, F1 scoring) are stored in results/figures and results/metrics respectively.

Optional non-comparable experiment (G1/G2 included): Decision Tree: Accuracy 0.873, Precision 0.735, Recall 0.962, F1 0.833, ROC-AUC 0.944; Random Forest: Accuracy 0.873, Precision 0.735, Recall 0.962, F1 0.833, ROC-AUC 0.937; Logistic Regression: Accuracy 0.873, Precision 0.750, Recall 0.923, F1 0.828, ROC-AUC 0.948 This is reported ONLY as an upper bound of what late-course grade information adds, not as an early-warning result.

13. Discussion

RQ1 (primary comparison): Random Forest achieved the highest F1 (positive class) and ROC-AUC among the three candidates, with the largest gain in precision (0.542); the linear and single-tree baselines landed on the same confusion matrix in this run - a coincidence possible on a 79-sample test split with a small grid, reported here exactly as measured rather than smoothed. CV on the training data gave best-CV-F1 values of Logistic Regression 0.554, Decision Tree 0.548, Random Forest 0.529 - i.e., within-noise differences between the algorithms on this dataset size, which is itself an important honest finding for a small-scale study: none of the three is a decisive

'winner', and any claims about superiority should be limited to descriptive reporting of this one split plus the CV pattern.

RQ3 (metric choice): for an early-warning use case, Recall of the at-risk class matters most (false negatives are a missed student), but operating with too low a Precision erodes teacher trust; the balanced choice is hence F1 (or an application-level threshold chosen on validation data, outside the scope of this project).

RQ2 (leakage quantification): including G1/G2 raises the best test accuracy/F1 from 0.696/0.520 to 0.873/0.833 - the difference is essentially the information those grades carry about the final exam, which early systems cannot use.

14. Limitations and Ethical Considerations

1. Statistical limits: 395 students from two schools, one subject, one country/era; a 79-student held-out split produces noisy metrics; single-split comparisons cannot establish algorithmic superiority more robust than the observed differences.

2. Scenario limits: 'features available early' is approximated by dataset attributes (e.g. absences/failures may be known only after early terms); a real deployment needs its own availability audit.

3. Definitions: the at-risk rule (G3 < 10) is a project-defined operational threshold on this grading scale, not a universal or clinical definition of academic risk.

4. Interpretation: all reported numbers are predictive associations. Absences, for example, are correlated with lower G3 but nothing here supports a causal claim; feature importance/rankings must not be read as causal or as a student judgment.

5. Ethics: attributes such as sex, family background, or parents' jobs can encode socioeconomic patterns; a real system would need explicit fairness auditing, transparency, student data protection, and human review before any educational decision. This project's artifacts are for teaching and demonstration only.

15. Conclusion and Future Work

Under a leakage-safe early-prediction protocol, the three classic algorithms performed in a close, statistically noisy band, with Random Forest giving the best measured F1 and ROC-AUC on the held-out split, and the comparison exposed the expected but large benefits of late-course grade information when it is wrongly allowed. Future work: cross-validated combined modelling of both subject files, hyperparameter search with repeated CV, cost-based thresholding, probability calibration, fairness/robustness audits, and semi-deployment studies with graded-feature availability.

16. References

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